

Q28

Momentum of electron *is* $p = mv$

$$\Delta p = m\Delta v = 9.11 \times 10^{-31} \times 500 \times 0.000100 = 4.56 \times 10^{-32} \text{ kg m s}^{-1}$$

Heisenberg's Uncertainty Principle:

$$\Delta p \Delta x \geq h . \text{ Calculating, } \Delta x \geq 0.015 \text{ m.}$$

Ans: C

Q29